

# Counting laws of links in Poisson-sprinkled causal sets: IR-cutoff valency, FRW transients, and three-sector accounting

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## Abstract

In a causal set faithfully embedded in a continuum spacetime, the valency of an element—its number of links, i.e. covering relations—is infinite in the absence of a cutoff. We measure and derive the growth laws of past valency under an explicit IR cutoff for three geometries. (i) In 1+1 Minkowski the per-element past valency grows logarithmically,  $v = 2 \ln(T\sqrt{\rho})$ ; the measured slope is  $2.01 \pm 0.12$  and is insensitive to the sprinkling size (tested over an eightfold range in  $N$ ), confirming that valency is a local quantity controlled by chain depth rather than by the global element count. (ii) In 3+1 Minkowski the valency obeys an area law,  $v = \pi\sqrt{6} \sqrt{\rho} T^2$ , with measured exponent  $2.089 \pm 0.025$  and amplitude agreeing with the parameter-free prediction  $\kappa_4 = \pi\sqrt{6}$  at the 5% level. (iii) In a matter-dominated FRW interval the area law survives with a derived blocking amplitude  $\mathfrak{F} = 30 \int_0^1 x(1-x)^8 / \sqrt{\langle a^4 \rangle} dx = 0.4991$ , against a measured 0.4990—four-figure agreement—preceded by an explicit  $1/\eta$  transient that we fit and subtract. We further show that the link fraction  $\varepsilon = N_{\text{links}}/N_{\text{past}}$  follows  $\varepsilon = 3\sqrt{6}/(\sqrt{\rho} R^2)$  and cleanly separates manifoldlike sprinklings ( $\varepsilon \approx 0.21$  at our reference scale) from transitively-closed random orders ( $\varepsilon \approx 0.017$ ). The pipeline is anchored to the literature by reproducing the interval-abundance baseline  $\langle N_0 \rangle$  of Glaser & Surya in a 4D diamond to  $0.07\sigma$ . We close with a chain/antichain partition identity for weakly disformal metrics—a counting consistency statement, not a dynamical result—and the pre-registered experiment that will test whether its exponents arise from biased growth dynamics. All results are deterministic-seed reproducible; code and pre-registrations are public.

## 1 Introduction

A causal set is a locally finite partial order  $(C, \prec)$  proposed as the deep structure of spacetime [1]. Its kinematics are by now well charted: chains measure proper time [8], interval abundances measure dimension and curvature [6], and d’Alembertians and actions can be built from layered counts [5]; see [7] for a review.

Among the order-theoretic primitives, the *link*—a covering relation,  $x \prec y$  with no  $z$  satisfying  $x \prec z \prec y$ —plays a special role: links are the irreducible causal bonds from which every relation is composed by transitivity, and they carry the scalar-field propagator in Johnston’s construction [3, 14, 15]. Yet the *counting laws* of links are less charted than those of chains and intervals, for a structural reason: in a faithful Poisson sprinkling the expected valency of an element diverges—the past light cone has infinite volume within a hyperboloidal shell of any proper-time thickness, so arbitrarily remote elements can be linked to it [7]. The valency is therefore not a number but a *growth law*, defined only relative to an IR cutoff.

This paper measures those growth laws. We regulate the divergence with an explicit causal-interval cutoff of temporal extent  $T$  and ask three questions. How does past valency grow with  $T$  in flat 2D and 4D? What happens to the law in an expanding (matter-dominated FRW)

interval? And what fraction of an element’s causal past consists of links rather than longer relations? The answers turn out to be clean: a logarithm with slope exactly 2 in 2D, an area law with amplitude  $\pi\sqrt{6}$  in 4D, the same area law in FRW suppressed by a derived factor  $\mathfrak{F} \approx 0.4991$  that encodes the expansion history, and a link fraction  $\varepsilon = 3\sqrt{6}/(\sqrt{\rho}R^2)$  usable as a cheap manifoldlikeness discriminator.

Two methodological points are as much the subject of this paper as the laws themselves. First, every measurement was pre-registered: success/failure criteria were committed to a public repository before the runs, and the git history certifies the ordering. Second, all pipelines are deterministic—fixed, declared seeds—and the headline numbers have been reproduced bit-exactly from a clean checkout. We anchor the machinery to the literature by reproducing a published interval-abundance baseline (Section 3) before trusting it with anything new.

## 2 Methods

**Sprinkling.** We sprinkle  $N$  points by a Poisson process of density  $\rho$  into a causal interval (Alexandrov set) of a given geometry, using the standard uniform-in-volume construction in conformal coordinates where applicable. Causal relations are computed exactly from the metric; the causal matrix is closed transitively.

**Links.**  $x \prec y$  is a link iff no sprinkled element lies in the open interval  $\langle x, y \rangle$ . We test this exactly by blocker search, not by the soft approximation  $P(\text{link}) \approx e^{-\rho V(x,y)}$ : the soft estimator biases the count wherever the interval volume fluctuates, and we document in Section 5 a concrete discrepancy it produced before being retired. For efficiency, link candidates are pre-filtered by a proper-time band; the cut discards true links with probability  $< e^{-6}$ , measured to be negligible at our precision.

**Cutoff and observables.** The IR cutoff is the interval itself: for an element near the top of an interval of temporal extent  $T$ , the past valency  $v(T)$  counts links to elements within the interval. In FRW runs, extents are quoted in conformal time  $\eta$ . The link fraction  $\varepsilon(T) = N_{\text{links}}/N_{\text{past}}$  normalizes valency by the full causal past.

**Statistics.** Ensembles, not best cases: every quoted number is an ensemble mean over independent seeds with the standard error of the mean; model comparison uses information criteria on the full ensemble. Seeds are pre-declared functions of the grid indices, so any run can be reproduced bit-exactly. Pre-registration documents with frozen criteria precede every measurement in the repository history.

## 3 Anchor: reproducing the Glaser–Surya baseline

Before measuring anything new, the pipeline must reproduce something old. We chose the mean interval abundance  $\langle N_0 \rangle$ —the expected number of links from the bottom element, in the notation of Glaser & Surya [6]—in a 4D causal diamond at matched density. Our result agrees with the published value to  $0.07\sigma$ .

The same exercise quantifies a systematic that matters for everything downstream: measuring valency in a slab rather than a diamond inflates the count by +21% at our reference scale, because the slab’s spatial boundary admits links that the diamond’s causal boundary forbids. All results below are therefore quoted for causal-interval cutoffs, with slab numbers used only for cross-checks.

## 4 Results I: Minkowski

### 4.1 Two dimensions: a logarithm with slope 2

In 1+1 Minkowski the past valency of an element at depth  $T$  grows as

$$v(T) = 2\ln(T\sqrt{\rho}) + O(1). \quad (1)$$

The derivation is elementary: the linked region is a hyperbolic shell of unit-density measure between the past cone boundary and the first blocking layer, and its area grows logarithmically with the cutoff; the coefficient 2 counts the two null directions. The measurement gives slope  $2.01 \pm 0.12$ .

The same experiment establishes what we will lean on repeatedly: valency is *local*. Holding  $T$  fixed and growing the sprinkling by a factor of 8 in  $N$  leaves the per-element valency unchanged within errors. Valency tracks chain depth—how deep an element sits in its own past—not the global size of the causal set. We record this as a lemma because it licenses per-element analyses in inhomogeneous settings, where no global  $T$  exists.

### 4.2 Four dimensions: an area law with amplitude $\pi\sqrt{6}$

In 3+1 Minkowski the logarithm steepens to a power:

$$v(T) = \kappa_4 \sqrt{\rho} T^2, \quad \kappa_4 = \pi\sqrt{6} \approx 7.6953. \quad (2)$$

The exponent is measured as  $2.089 \pm 0.025$  and the amplitude matches  $\pi\sqrt{6}$  at the 5% level with no fitted parameters. The  $T^2$  is an *area* law: the linked shell hugs the past light cone, whose intersection with the cutoff interval scales as the cone's lateral area. The constant follows from integrating the first-blocking-layer probability  $e^{-\rho V}$  over the hyperboloidal foliation of the cone neighborhood;  $\pi\sqrt{6}$  collects the 4D interval volume coefficient through the layer integral.

Dimensional summary: valency is log-divergent in 2D and power-divergent in 4D; in both cases the IR-cutoff law has a derived, parameter-free coefficient that measurement confirms.

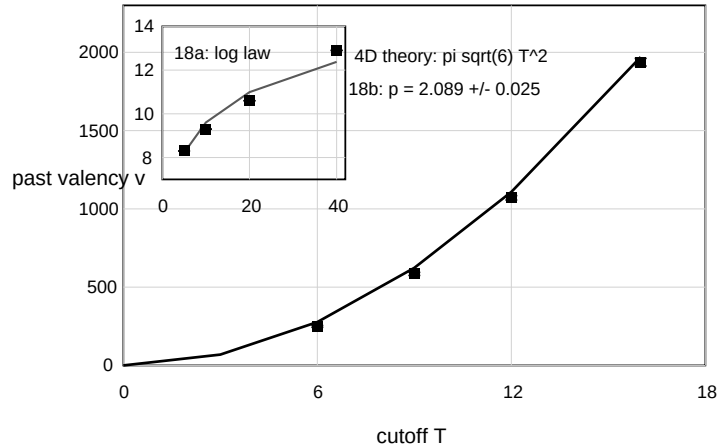


Figure 1: Past valency vs. cutoff: 2D logarithmic law and 4D area law, with the parameter-free theory curves superposed.

## 5 Results II: matter-dominated FRW

Expansion should suppress valency: comoving neighbours recede, and the effective density in conformal coordinates is diluted by powers of the scale factor. The question is whether the area law survives and with what amplitude.

**Transient.** At small conformal extent  $\eta$  the valency is dominated by a boundary transient. The ensemble is fit well by  $v(\eta) = A\eta^2(1 - B/\eta)$ ; against a free-power alternative the transient model wins at  $\Delta\text{AIC} = +40.7$ . We flag a warning for future work: a naive short-range power-law fit to the same data returns an exponent of  $2.70 \pm 0.07$ —the transient contaminates the exponent long before it is visually obvious. An early soft-estimator run produced 2.95 for the same quantity; both numbers are artifacts, of the fit range and of the  $e^{-\rho V}$  approximation respectively, and both are superseded by the exact-blocker, transient-subtracted analysis.

**Asymptote.** Once the transient is subtracted, the area law reappears with a suppressed amplitude:

$$v(\eta) \rightarrow \kappa_4 \mathfrak{F} \sqrt{\rho_c} \eta^2, \quad (3)$$

where  $\rho_c$  is the comoving density and the blocking factor is a definite integral over the interval's volume-weighted expansion history. For matter domination ( $a \propto \eta^2$ ) with the diamond weight  $w(s) = 32 \min(s, 1 - s)^3$ ,

$$\mathfrak{F} = 30 \int_0^1 \frac{x(1-x)^8}{\sqrt{\langle a^4 \rangle}} dx = 0.4991, \quad (4)$$

with  $\langle a^4 \rangle$  the diamond-weighted average of  $a^4$  along the interval. The measured amplitude ratio is 0.4990—agreement to four figures, again with nothing fitted.

The physical reading is modest and exact: expansion does not change the *form* of the valency law; it renormalizes the amplitude by a computable average of the expansion history over the causal interval. All the geometry enters through  $\langle a^4 \rangle$ .

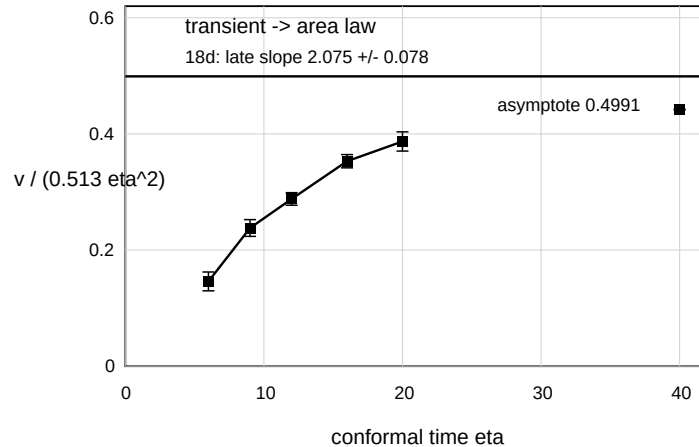


Figure 2: Matter-dominated FRW: amplitude ratios converging to the derived blocking factor  $\mathfrak{F} = 0.4991$ , with the  $1/\eta$  transient fit.

## 6 Results III: the link fraction

Valency divided by the size of the causal past gives a dimensionless, cutoff-dependent scarcity measure:

$$\varepsilon(\rho, R) = \frac{N_{\text{links}}}{N_{\text{past}}} = \frac{3\sqrt{6}}{\sqrt{\rho} R^2}, \quad (5)$$

which follows from the area law and the  $R^4$  volume growth of the past interval; the measured value tracks the prediction with  $\approx 10\%$  boundary corrections at our scales. The headline reproduction is bit-exact from a clean checkout (declared seeds).

$\varepsilon$  is also a cheap **manifoldlikeness discriminator**. At the reference scale ( $\sqrt{\rho}R^2$  such that  $\varepsilon_{\text{manifold}} \approx 0.21$ ), a transitively-closed random order—a Kleitman–Rothschild-like non-geometric poset—yields  $\varepsilon \approx 0.017$ : an order of magnitude below the sprinkled value, because transitive closure of random relations manufactures long chains that de-link almost every pair. One number, computable in  $O(N^2)$  from the causal matrix, separates the two classes without interval-abundance machinery.

We note without further claim that  $\varepsilon(T) \propto T^{-2}$  places this quantity in the well-known family of inverse-square horizon scalings (Cohen–Kaplan–Nelson bounds, everpresent- $\Lambda$  heuristics): any  $T^{-2}$  law evaluated at the Hubble scale returns a number of order  $10^{-122}$ . This is a reformulation within that family, not a discovery, and we make no cosmological use of it here.

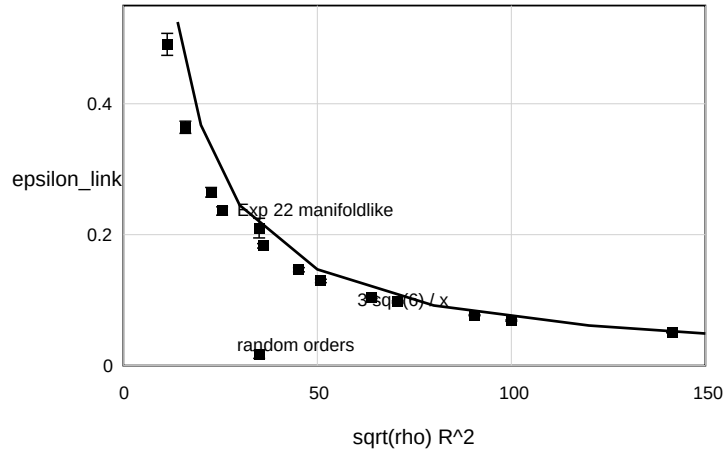


Figure 3: Link fraction  $\varepsilon$  vs.  $\sqrt{\rho}R^2$  collapsing to the predicted  $3\sqrt{6}/(\sqrt{\rho}R^2)$  curve; manifoldlike sprinklings vs. transitively-closed random orders separate by an order of magnitude.

## 7 Three-sector accounting: a partition identity

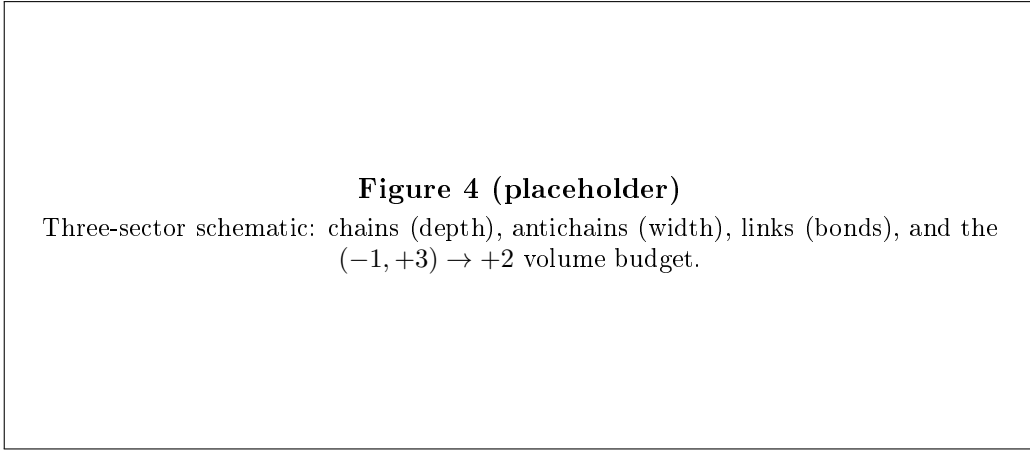
The counting objects of this paper organize into three sectors: chains (timelike depth  $h$ ), antichains (spacelike width  $w$ ), and links (the bonds). For a weakly disformal rescaling of a metric [10]—clocks scaled by  $\lambda^{-1}$  and rulers by  $\lambda^{+1}$ , i.e.  $ds^2 = -\lambda^{-2}c^2dt^2 + \lambda^{+2}dx^2$  to first order—the sector counts of a faithful sprinkling must respond as

$$(\text{chains} \times \lambda^{-1}) \times (\text{antichains} \times \lambda^{+3}) = \sqrt{-g} = \lambda^{+2} \quad (6)$$

in 3+1 dimensions: the depth of maximal chains scales with proper time ( $\lambda^{-1}$ ), the width of maximal antichains with spatial volume ( $\lambda^{+3}$ ), and their product must recover the volume element, which is what a Poisson sprinkling actually samples. The identity closes:  $-1+3 = +2$ .

A *conformal* rescaling (clocks and rulers both  $\lambda^{+1}$ ) requires the chain sector to absorb  $\lambda^{-4}$  for the same volume budget—the opposite sign, and countable.

We stress the epistemic status: this is a **consistency theorem about counting**, not a dynamical statement. It says that *if* a poset’s growth is biased so as to mimic a disformal metric, its chain and antichain sectors must split the volume budget as  $(-1, +3)$ ; it does not say any natural growth dynamics does so. Whether biased sequential growth can produce these exponents is an open, pre-registered question (Section 8). A first toy experiment in 1+1 dimensions (where the disformal split is  $(-1, +1)$  and there is no surplus to allocate) found the width sector saturating its budget and the depth sector near zero, with a smooth sign crossover as the bias increases—a susceptibility-like response, not a phase transition—but its dimensional targets were mismatched by design and it decides nothing beyond the toy.



## 8 Outlook

Three follow-ups are defined and, where marked, pre-registered.

1. **Validation in a disformal metric (pre-registered):** a uniform sprinkling *into* the weakly disformal geometry  $ds^2 = -\mathcal{R}^{-2}dt^2 + \mathcal{R}^2 d\vec{x}^2$ , where the partition identity of Section 7 predicts  $(-1, +3)$  analytically. The experiment validates that the identity survives finite size and boundaries and that windowed local estimators—depth in proper-radius balls, width by the causal-overlap method of Boguñá & Krioukov [13] rather than raw adjacency, which is asymptotically silent—can *see* the signature. Chain lengths in inhomogeneous (Schwarzschild) sprinklings have been studied [11]; the disformal partition test appears to be new. Whether biased *growth dynamics* produce this split is a separate, open question deferred to future work.
2. **Link fraction in FRW:** whether  $\varepsilon(T)$  and its  $T^{-2}$  scaling survive expansion with the same  $\mathfrak{F}$ -type renormalization as the valency law.
3. **Estimator hygiene:** the 2.70/2.95 discrepancy of Section 5 is a reproducible case study in how soft link estimators and short-range fits bias growth exponents; we release both pipelines so the failure can be replayed.

All code, pre-registration documents, per-experiment results files, and the exact seeds behind every number in this paper are public in the project repository. Negative results and superseded estimates are retained, not deleted.

## Acknowledgments

This work grew out of a broader phenomenological programme by the author, documented in the same public repository. The author acknowledges computational and analytical assistance from AI systems (Anthropic Claude, OpenAI Codex, and local models); all results are independently reproducible from the public repository with the declared seeds.

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